

Apply First Normal Form (1NF) and Second Normal Form (2NF)

1. First Normal Form (1NF)

A relation is in **First Normal Form (1NF)** when:

- Each column contains **atomic (single) values**.
- There are no repeating groups or multivalued attributes.
- Each record can be uniquely identified using a **Primary Key**.

In the E-Commerce database, the entities are separated into individual tables such as Customer, Seller, Category, Product, Orders, Order_Details, Payment, and Shipping. Each table uses a primary key to uniquely identify its records.

Schema in 1NF

```
CUSTOMER(  
    Customer_ID PK,  
    Customer_Name,  
    Email,  
    Phone,  
    Address,  
    Registration_Date  
)
```

```
SELLER(  
    Seller_ID PK,  
    Seller_Name,  
    Email,  
    Phone,  
    Address  
)
```

```
CATEGORY(  
    Category_ID PK,
```

Category_Name
)

PRODUCT(
Product_ID PK,
Product_Name,
Category_ID FK,
Seller_ID FK,
Price,
Stock_Quantity,
Product_Status
)

ORDERS(
Order_ID PK,
Customer_ID FK,
Order_Date,
Order_Status,
Total_Amount
)

ORDER_DETAILS(
Order_Detail_ID PK,
Order_ID FK,
Product_ID FK,
Quantity,
Unit_Price,
Subtotal

)

PAYMENT(

Payment_ID PK,

Order_ID FK,

Payment_Method,

Amount,

Payment_Date,

Payment_Status

)

SHIPPING(

Shipping_ID PK,

Order_ID FK,

Shipping_Address,

Tracking_Number,

Delivery_Status

)

Thus, all attributes contain single values, and repeating product information for an order is handled using the separate ORDER_DETAILS table.

2. Second Normal Form (2NF)

A relation is in **Second Normal Form (2NF)** when:

1. It is already in **1NF**.
2. Every non-key attribute is fully dependent on the **entire Primary Key**.
3. There are no **partial dependencies**.

In this database design, most tables have a single-column primary key. Therefore, partial dependency does not occur because every non-key attribute depends on the complete primary key.

2NF Analysis

Table	Primary Key	Dependency
CUSTOMER	Customer_ID	Customer_ID → Customer_Name, Email, Phone, Address
SELLER	Seller_ID	Seller_ID → Seller_Name, Email, Phone, Address
CATEGORY	Category_ID	Category_ID → Category_Name
PRODUCT	Product_ID	Product_ID → Product_Name, Price, Stock_Quantity, Product_Status
ORDERS	Order_ID	Order_ID → Order_Date, Order_Status, Total_Amount
ORDER_DETAILS	Order_Detail_ID	Order_Detail_ID → Order_ID, Product_ID, Quantity, Unit_Price, Subtotal
PAYMENT	Payment_ID	Payment_ID → Payment_Method, Amount, Payment_Date, Payment_Status
SHIPPING	Shipping_ID	Shipping_ID → Shipping_Address, Tracking_Number, Delivery_Status

The use of Order_Detail_ID as the primary key in ORDER_DETAILS ensures that all its attributes are dependent on one complete key. The foreign keys Order_ID and Product_ID establish relationships without creating partial dependencies.

Final Normalized Structure

1NF

|

└─ Atomic values

└─ No repeating groups

└─ Primary Key for each table

|



2NF

|

└─ Already satisfies 1NF

└─ No partial dependency

└─ All non-key attributes fully depend

on their respective Primary Key

Conclusion

The E-Commerce Order Management database satisfies **1NF and 2NF** because the data is divided into separate relations, attributes contain atomic values, and non-key attributes are fully dependent on their respective primary keys. This structure helps reduce redundancy and prevents insertion, update, and deletion anomalies.